Soumil Iyer

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EDUCATION

Purdue University, West Lafayette, IN

Master of Science in Mechanical Engineering (GPA: 4.00) — January 2026 - December 2026 (Expected)

Bachelor of Science in Mechanical Engineering (GPA: 3.38) — August 2022 - December 2025

EXPERIENCE

Rapid Prototyping Lab Student Employee | Purdue University — **August 2026 - Present**

Design for Additive Manufacturing | CAD & DFM Review | Rapid Prototype Support

- Review CAD and STL models for FDM feasibility, identifying thin walls, unsupported geometry, fit risks, and design revisions before printing
- Select materials, print orientation, supports, infill, and Bambu Studio or Cura settings to balance part strength, fit, finish, and build reliability
- Diagnose print failures involving CAD geometry, slicer settings, materials, and machine calibration; recommend corrective process changes
- Set up, calibrate, and maintain FDM printers while guiding iterative CAD-to-part workflows and repeatable prototype fabrication for varied parts
- Explain how CAD geometry and print parameters influence build time, dimensional accuracy, strength, and reliability during design reviews

Graduate Teaching Assistant - Intro to Manufacturing | Purdue University — **January 2026 - May 2026**

Engineering Drawings & BOM Review | GD&T & Metrology | Design-to-Build Support

- Reviewed CAD drawings, BOMs, and assembly plans in a 19-student class section to identify tolerance, fit, and DFM risks before fabrication
- Applied ASME Y14.5 GD&T and dimensional inspection to assess drawing requirements, mating part interfaces, and manufactured assembly fit
- Guided CNC, manual machining, and assembly work; investigated dimensional defects, process setups, and design-to-hardware discrepancies
- Compared finished components against design requirements and coached drawing, tolerance, and assembly revisions through final inspection

M.E. Undergraduate Research Intern | Purdue University — **June 2025 - August 2025**

Materials Characterization | ASTM D638 & Ultrasonic NDE | Test & Analysis

- Planned and executed a material study using 60+ tensile samples and 24 ultrasonic disks across six thermoplastics and two FDM orientations
- Characterized stiffness, anisotropy, and failure using ASTM D638 tensile testing and ultrasonic NDE to compare material and build responses
- Converted stress-strain and ultrasonic data in MATLAB into Young's modulus and wave-speed metrics for material and orientation comparison
- Documented specimen controls, test procedures, uncertainty, and orientation-dependent failure findings in a report on FDM material behavior

M.E. Undergraduate Research Assistant - Energy Storage Analytics | Purdue University — **August 2024 - December 2024**

Engineering Data Visualization | Python GUI Development | Li-ion Performance Analysis | Cell Design Fundamentals

- Developed an interactive Python application using customtkinter to import, process, and visualize 1,000+ Li-ion battery performance data points
- Created reusable battery-comparison plots and data-processing views to reveal cell-performance trends and standardize technical analysis records
- Compared Li-ion performance datasets to relate measured cell behavior to battery-design tradeoffs and identify questions for further analysis

Teaching Assistant - Thermodynamics I & ENGR 131 | Purdue University — **August 2024 - December 2024**

PROJECTS

Mechanical Engineering Capstone Project - Assisted Eating Utensil — **August 2025 - December 2025**

- Led CAD design and FDM prototype iterations for a tremor-assist utensil, translating user needs into performance and <$25 cost targets
- Benchmarked weighted, powered, and passive stabilization concepts; selected a bearing mechanism and assessed risks at PDR & CDR reviews
- Translated senior-living user feedback into an enlarged ribbed grip, hand & wrist straps, rotation-stop feature, and detachable steel spoon head
- Iterated FDM handle geometry, bearing fits, and detachable stainless-steel utensil interfaces to improve printability, cleanability, and assembly fit
- Built a costed parts list around an FDM handle, bearing, and steel hardware, maintaining ~$13 material cost against a sub-$25 price target
- Validated the utensil with 2 m drops, 14 dishwasher cycles, and 7/3 lbf insertion/removal force tests; documented test limitations and design risks

Wind Tunnel Aerodynamic Analysis - Race Car Spoiler | ME 30801 — **March 2025 - April 2025**

- Designed a SolidWorks race-car spoiler, fabricated an FDM test article, and calibrated load-cell/LVDT instrumentation before wind-tunnel trials
- Executed 3 wind-tunnel trials per 0-60 deg angle at 30.88 m/s, recording 15-second measurement intervals for repeatable force comparison
- Converted raw sensor output into lift and drag coefficients in MATLAB; compared repeated measurements using averages and uncertainty bars
- Identified peak Cl ~0.65 near 40 deg before flow separation and documented a ~2 mm print shift as a potential source of measurement error

Mechanical Engineering Design Project - Foldable Travel Hanger — **January 2024 - April 2024**

- Led CAD design, 3D printing, and prototyping for a lightweight, foldable travel hanger within a four-person team, from concept to assembly
- Translated portability, garment-clearance, and durability needs into measurable requirements; benchmarked alternative folding hanger concepts
- Modeled hinge components and folding assembly in Siemens NX, defining geometry, interfaces, and folding clearances before prototyping
- FDM-printed components and assembled a working prototype to assess hinge travel, part fit, folding sequence, portability, and feasibility
- Evaluated prototype motion and component fit against CAD intent, identifying assembly constraints and opportunities to refine the mechanism

Little Blazer Engine - Parametric CAD & Assembly | CGT 163 — **March 2023 - April 2023**

- Recreated engine parts from dimensioned drawings in Siemens NX, using parametric features to preserve component geometry and design intent
- Built a multi-part engine assembly with geometric constraints; checked component positioning, mating surfaces, alignment, and fit in NX

SKILLS & CERTIFICATIONS

CAD & Mechanical Design: SolidWorks, Siemens NX (PLM/PDM), CATIA, Fusion 360, AutoCAD, parametric modeling, parts/assemblies

Product Development & DFM: ASME Y14.5 GD&T, tolerance stack-up, DFM/DFA, BOMs, drawings, design reviews, material selection

Prototyping & Manufacturing: FDM/FFF, Bambu Studio, Cura, CNC/manual milling and turning, mechanical assembly, metrology

Mechanical Analysis: Fusion 360 FEA, static analysis, shape optimization, mechanics of materials, solid mechanics, heat transfer

Testing & Validation: ASTM D638, ultrasonic NDE, LabVIEW, DAQ, LVDTs/load cells, sensor calibration, uncertainty analysis

Statistics & Software: MATLAB, Python, Minitab, DOE/ANOVA, Excel, Simulink

Additional Engineering Tools: Studio 5000, Ladder Logic, ROBOGUIDE, PID/PI, root locus, LTspice, Arduino

Certifications: SOLIDWORKS Purdue Milestones (2026), CSWA In progress; Purdue FEA (2025); Simulink, MATLAB, Python, Arduino